

veloped, but their basic functioning is based on the methods described above.

High frequency methods

For computing field quantities of Maxwell equation at electrically large structures, optical methods have been developed. These methods can be divided into (a) field-based methods like UTD (uniform theory of diffraction) and (b) current-based methods like PO (physical optics).

In PO method the current distribution is calculated by the use of Green's function (MOM) and then this distribution is applied to the boundary conditions of the structure. Then calculations are performed on the structure by inducing that current distribution and updating the main results. The incident wave on the structure continues or passes through it as if the structure was not present.

The UTD method employs a different strategy. Here also the current

distribution is applied to the boundary condition but the structure blocks the incident waves on each part of the structure. The field quantities are then determined from the diffracted waves. Consider a reflector antenna placed in front of it. The pattern and quantities will be calculated by PO in the front region while the backside region will be computed by UTD.

CST, HSS, FEKO solver overview

Computer simulation technology microwave studio (CST MWS) is one of the most comprehensive tools for RF designers and engineers. It provides the maximum of six solvers, namely:

1. Time domain solver. It uses hexahedral mesh and is very efficient for high frequencies. It provides two alternative methods: transient solver (FIT) and TLM solver (for better simulation of EMC/EMI models). It calcu-

lates transmission of energy between various ports and open structures of the model. Typical applications include TDR, RCS and dispersive materials.

2. Frequency domain solver. It uses hexahedral and tetrahedral mesh.

3. Eigen mode solver. It is recommended strongly for loss-free structures.

4. Integral solver. It uses MLFMM technique.

5. Multilayer solver. For planar structures as it is based on MOM.

6. Asymptotic solver. It uses optical techniques for electrically large structures.

Apart from CST, HFSS and FEKO are also widely used softwares for CEM applications. HFSS provides two solvers: FEM and HFSS-IE (MLFMM based). FEKO provides five solvers: FEM, MOM, PO, UTD and MLFMM. ●

The author, a B.Tech in EC, has contributed some other interesting articles as well in the past

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